of chemistry is a matter of applied quantum mechanics. To understand how the sun shines, or why tables are solid, you need quantum mechanics.

Imagine closing your eyes. Hopefully things look pretty dark. You might think that makes sense, because no light is coming in. But that's not quite right; infrared light, with a slightly longer wavelength than visible light, is being emitted all the time by any warm object, and that includes your own body. If our eyes were as sensitive to infrared light as they are to visible light, we would be blinded even when our lids were closed, from all the light emitted by our eyeballs themselves. But the rods and cones that act as light receptors in our eyes are cleverly sensitive to visible light, not infrared. How do they manage that? Ultimately, the answer comes down to quantum mechanics.

Quantum mechanics isn't magic. It is the deepest, most comprehensive view of reality we have. As far as we currently know, quantum mechanics isn't just an approximation of the truth; it is the truth. That's subject to change in the face of unexpected experimental results, but we've seen no hints of any such surprises thus far. The development of quantum mechanics in the early years of the twentieth century, involving names like Planck, Einstein, Bohr, Heisenberg, Schrödinger, and Dirac, left us by 1927 with a mature understanding that is surely one of the greatest intellectual accomplishments in human history. We have every reason to be proud.

On the other hand, in the memorable words of Richard Feynman, "I think I can safely say that nobody understands quantum mechanics." We *use* quantum mechanics to design new technologies and predict the outcomes of experiments. But honest physicists admit that we don't truly *understand* quantum mechanics. We have a recipe that we can safely apply in certain prescribed situations, and which returns mind-bogglingly precise predictions that have been triumphantly vindicated by the data. But if you want to dig deeper and ask what is really going on, we simply don't know. Physicists tend to treat quantum mechanics like a mindless robot they rely on to perform certain tasks, not as a beloved friend they care about on a personal level.

This attitude among the professionals seeps into how quantum mechanics gets explained to the wider world. What we would like to do is to present a fully formed picture of Nature, but we can't quite do that, since physicists don't agree about what quantum mechanics actually says. Instead, popular treatments tend to emphasize that quantum mechanics is